Esch384 Hash on FPGA: Design Document

ECE 545 Project, Phase 2 • Henry Su, Nick Ngo • November 10, 2026

Goal

Build Esch384 in hardware on an FPGA. Esch384 is a hash function from the SPARKLE family, a finalist in NIST's Lightweight Cryptography competition. It takes a message of any length and gives a 384-bit (48-byte) digest. It is a *sponge*: the message goes in 16 bytes at a time (rate 128 bits), and the rest of the 512-bit state (capacity 384 bits) is never touched directly.

Design strategy

We build a simple *iterative* design first: one permutation step per clock cycle. We make it fully correct before we try to make it smaller. Hashing a message of n blocks takes $8n + 20$ steps (8 per block, 4 extra on the last block, and 16 to output the 3 digest pieces).

Components

Block	What it does
Alzette	Mixing box on two 32-bit words: add, rotate, XOR (4 rounds). 8 copies, one per branch.
Linear layer	Mixes the 8 branches together and swaps their order.
SPARKLE512	One step = constant + 8 Alzettes + linear layer; repeated 8 or 12 times.
Datapath	512-bit state register, message padding and injection, digest output.
Controller	State machine: take block, run steps, output 3 digest pieces; valid/ready handshake.
Top level	Datapath + controller, tested against all known-answer tests.

Tools and libraries

SystemVerilog (kept close to Verilog-2001); Xilinx Vivado 2026.1 for synthesis and xsim for simulation (Questa as backup); target board Arty A7 (Artix-7, `xc7a100tcsg324-1`). Reference: the official NIST C code for Esch384, plus Python scripts that create test vectors and compare outputs. No outside hardware libraries (IP cores). Code is on GitHub.

Optimization target: area

Lightweight cryptography is about small area, so area is our target. We will report LUTs, flip-flops, maximum clock speed, latency and throughput for every version.

Planned optimizations

- **Serialize the Alzettes:** use 4, 2 or 1 Alzette box instead of 8, and take more cycles per step. This is the same trick as the round-serial AES designs from lecture 13.
- Share adders, reuse registers, and try a narrower datapath.
- Each idea gets its own git branch. All tests are re-run after every change.

Risk management

- **Wrong results:** every block is tested against the official C code on its own before it is used. We also break a copy on purpose to make sure the test catches it.
- **Hard-to-find bugs:** the reference model prints the state after every step, so we can find the first step where the hardware and the reference differ.
- **Schedule:** tasks are small and split between us. The working version gets a git tag before any optimization, so we can always go back to it.

Progress so far

Item	Status	Result
Setup: tools, repo, course notes	Done	Vivado/xsim 2026.1 working
Reference model and test vectors	Done	Reproduces all 1025 official test vectors
Alzette	Done	64/64 tests pass, 254 LUTs
Linear layer	Done	64/64 tests pass, 320 LUTs
SPARKLE512 permutation	Done	71/71 tests pass, 2023 LUTs, 521 FFs; 9 / 13 cycles
Datapath, controller, top level	Next (Nick)	Specs and test plan written
Synthesis and area optimization	Planned	–

Challenges and how we solved them

- **Byte and word order were unclear in the spec.** We checked them against the official C code.
- **The simulator (xsim) crashed** on one testbench line. We rewrote the line a simpler way.
- **No admin rights and a full home drive** on the lab machines. We installed packages locally and cleaned old files.
- **No project handout or AI-use rules found.** We log all AI tool use and will ask at office hours.